What IAT papers ask you to do

A visual guide for Class 12 students | Nine acquired English papers, 2017–2026

The main lesson is to prepare for **reasoning across all four sciences**. Repeated topics deserve extra practice, but a chapter with fewer recent questions is still part of your syllabus.

Four findings to use in your preparation

• **Biology:** genetics/evolution and human physiology account for 22 of the 58 recent current-scope questions. Practise mechanisms, crosses and observations, not only definitions.

• **Chemistry:** bonding, organic principles, coordination and carbonyl/carboxyl chemistry together account for 23 of 55 recent current-scope questions. Connect structures with reactions and reasons.

• **Mathematics:** coordinate geometry, differentiation and integration each contribute 6 of 59 recent current-scope questions. Questions also test domains, cases and branch choices.

• **Physics:** field/potential reasoning and work/energy lead this sample. Choosing a model, reading a graph and keeping signs correct often matter before arithmetic.

How to read the numbers

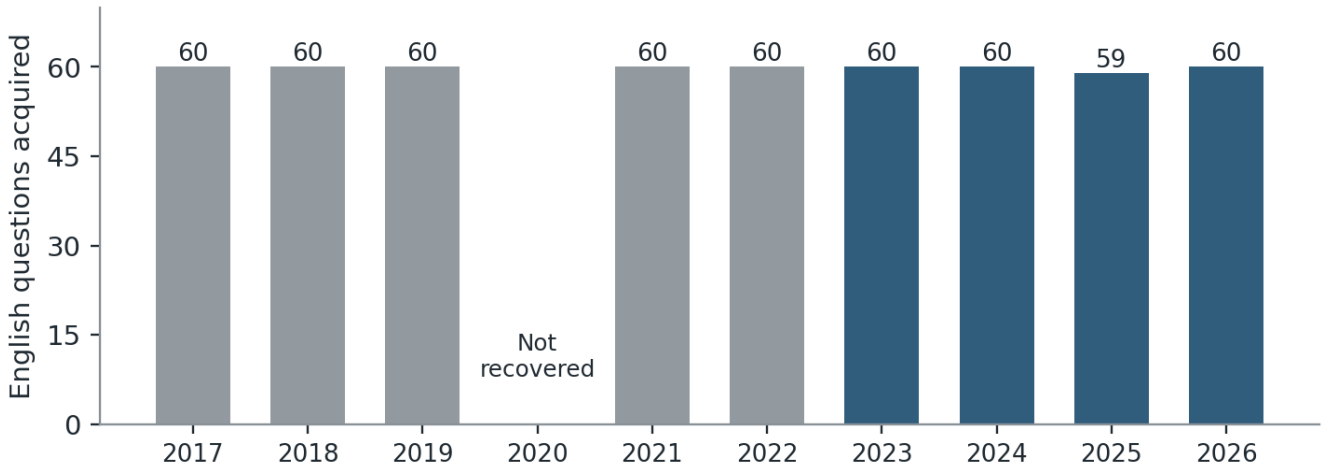
A weightage is a share of the questions we counted. For example, 12 genetics questions out of 58 recent Biology questions gives 20.7%. It does **not** promise that the next paper will repeat this share.

Where to go	What you will find
Pages 2–3	The papers included, exam pattern and scoring
Pages 4–8	Every retained chapter family, shown subject by subject
Pages 9–12	Repeated areas, question styles, difficulty and syllabus changes
Pages 13–14	Study priorities, an error log and sources

The verified exam rules here are for 2026. This report supports 2027 preparation, but does not claim a verified 2027 notification. Sources and limits are listed on page 14.

Which papers are included?

Count the evidence before treating a pattern as reliable.



English question counts from the acquired archive [S1–S2]. Grey bars are older papers; blue bars are the recent comparison window. No bar is drawn for the unrecovered 2020 paper.

Set	Questions	Why it matters
All acquired English papers	539	Nine papers; useful for breadth and style
Recent papers: 2023–2026	239	Closer to the retained syllabus; four papers only
Recent, current-scope families	231	58 Biology + 55 Chemistry + 59 Maths + 59 Physics

Three gaps to remember

• The acquired 2025 English copy has 59 labelled questions, ending at Physics Q14. We did not invent a replacement.

• The 2026 source is a provisional answer-key document. Its first-option answer convention is not evidence about live-exam answer letters.

• Hindi copies support access but are not counted again. Mirror sources and OCR errors are recorded; check the original page if a formula looks wrong.

Know the pattern and count marks correctly

Verified IAT 2026 rules [S3]. Recheck the target-year notification before the exam.

Rule	What it means
60 single-correct MCQs	15 each in Biology, Chemistry, Mathematics and Physics
180 minutes	An overall average of 3 minutes per question; individual items can take less or more
+4 / -1 / 0	Correct / incorrect / unanswered
240 maximum marks	All four subjects contribute; there is no best-three rule
Calculator	On-screen non-programmable scientific calculator; personal calculators are not allowed

A simple score example

Suppose you get **35 correct, 15 wrong and leave 10 unanswered**. Your score is (35 × 4) - (15 × 1) = **125 out of 240**. This is an illustration, not a result measured from the PYQs.

Use the time for decisions, not just calculations

• Start with questions where you can identify a method. A long stem is not automatically hard, and a short stem is not automatically easy.

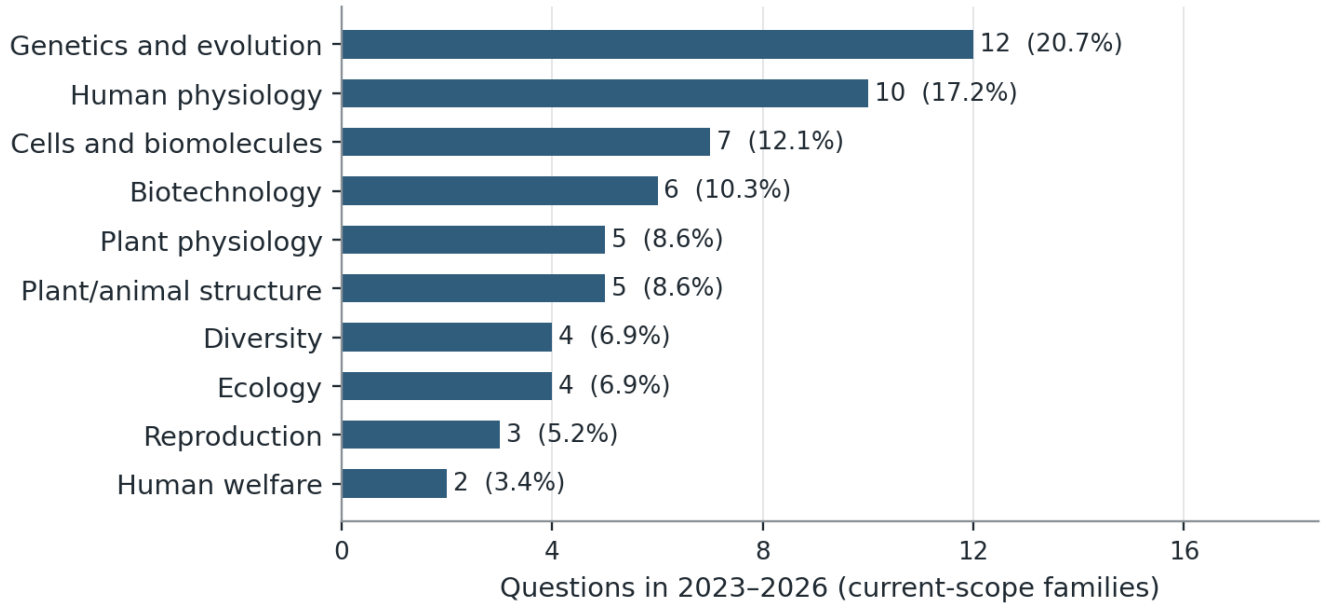
• For a graph, read both axes and signs. For a reaction, track each functional group. For a probability, check what information is already given.

• If you cannot see a productive next step, mark the question for a later pass. Return with time to check units, cases and option wording.

No score-to-rank estimate, cut-off forecast or fixed attempt target is justified by this chapter-count study. Actual scores depend on the paper and the candidate group.

Biology: genetics and physiology recur

Current-scope recent sample: 58 questions. Broad units are not all the same size.



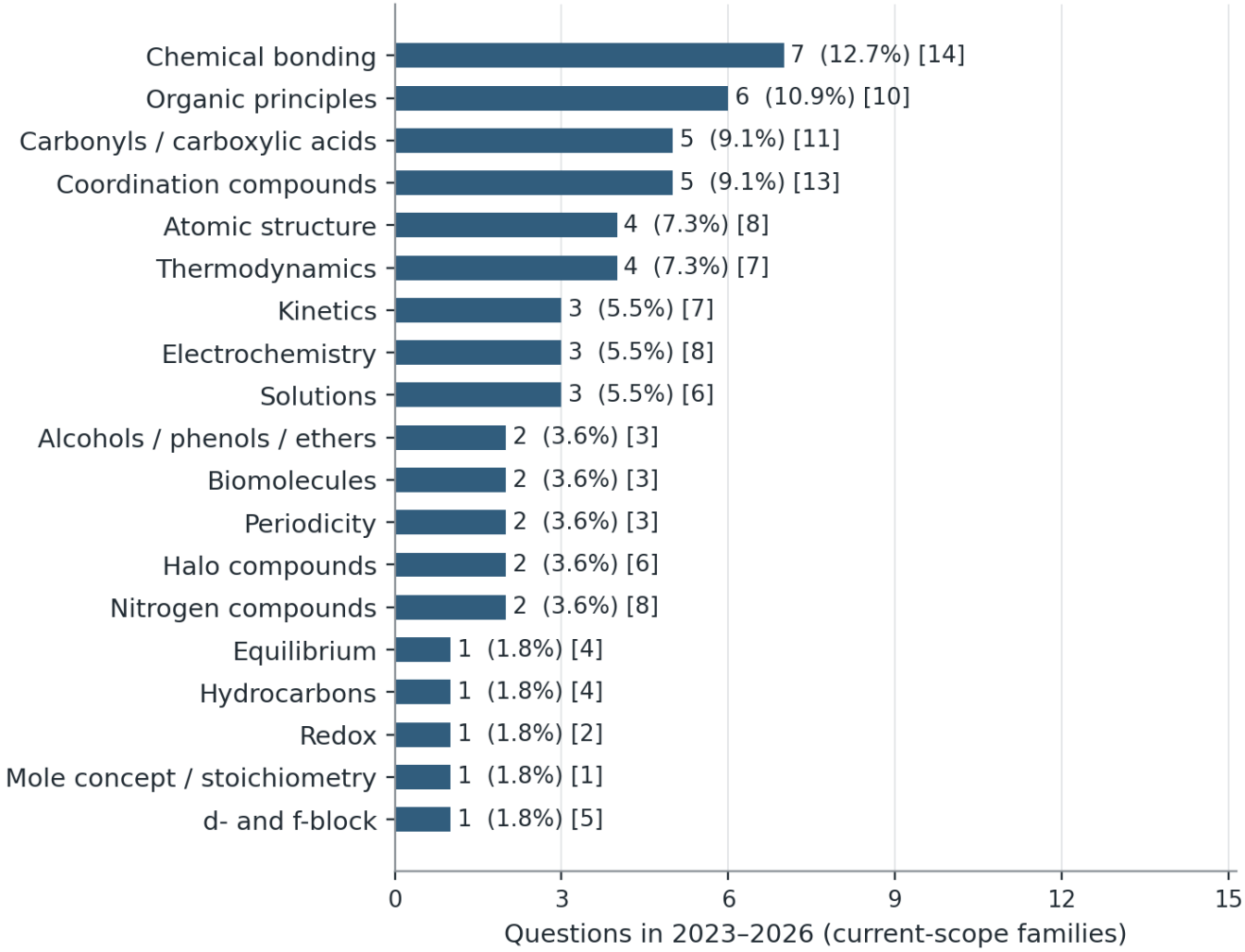
Source: manual PYQ chapter ledger [S2]. Recent = 2023–2026. Shares use 58 current-scope questions in Biology; each question has one primary chapter.

Family	All years	Recent
Diversity	10	4
Plant/animal structure	11	5
Cells and biomolecules	16	7
Plant physiology	11	5
Human physiology	21	10
Reproduction	5	3
Genetics and evolution	25	12
Human welfare	8	2
Biotechnology	13	6
Ecology	10	4

Practise: inheritance probabilities, cell processes, hormone effects and experimental controls. In a cross, distinguish a gamete frequency from an offspring probability.

Chemistry: connect structure, reason and reaction

Current-scope recent sample: 55 questions. The chart lists all 19 retained families.



Source: manual PYQ chapter ledger [S2]. Recent = 2023–2026. Shares use 55 current-scope questions in Chemistry; each question has one primary chapter. Bracketed counts [ ] show all acquired years for that family.

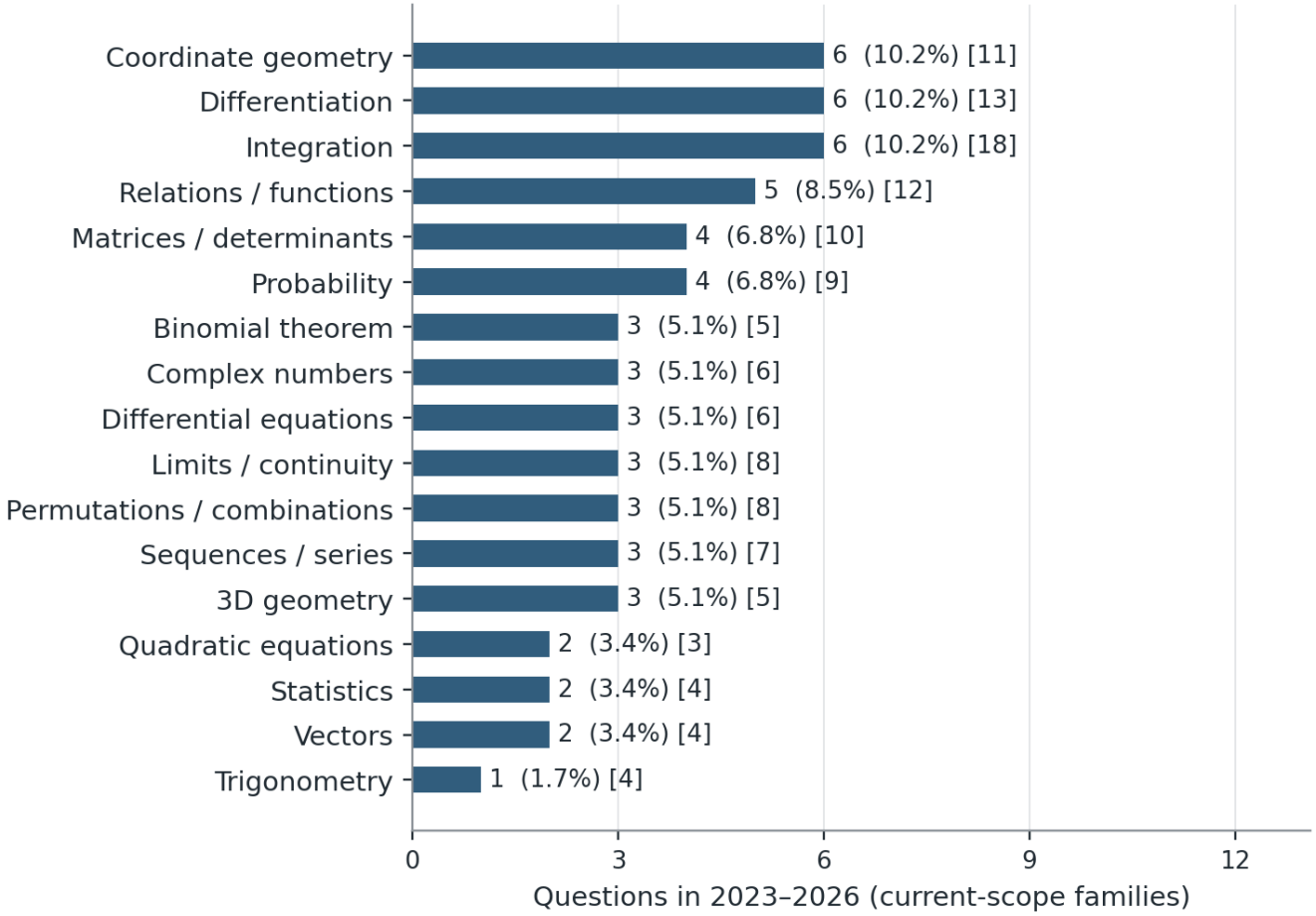
The main cluster: bonding 7; organic principles 6; coordination 5; carbonyl/carboxyl chemistry 5. These are 23 of 55 questions, or 41.8%. Other families together still contribute 58.2%.

Practise: interpreting molecular structures, comparing reagents and using equilibrium before a calculation. Physical chemistry remains important even when its questions are split across several chapters.

All-year / recent counts for the remaining detail are preserved in the companion chapter-count appendix [S2]. Historical topics such as solid state and surface chemistry are not silently included in this current-scope chart.

Mathematics: calculus, functions and geometry

Current-scope recent sample: 59 questions. A single method will not cover this range.



Source: manual PYQ chapter ledger [S2]. Recent = 2023–2026. Shares use 59 current-scope questions in Mathematics; each question has one primary chapter. Bracketed counts [ ] show all acquired years for that family.

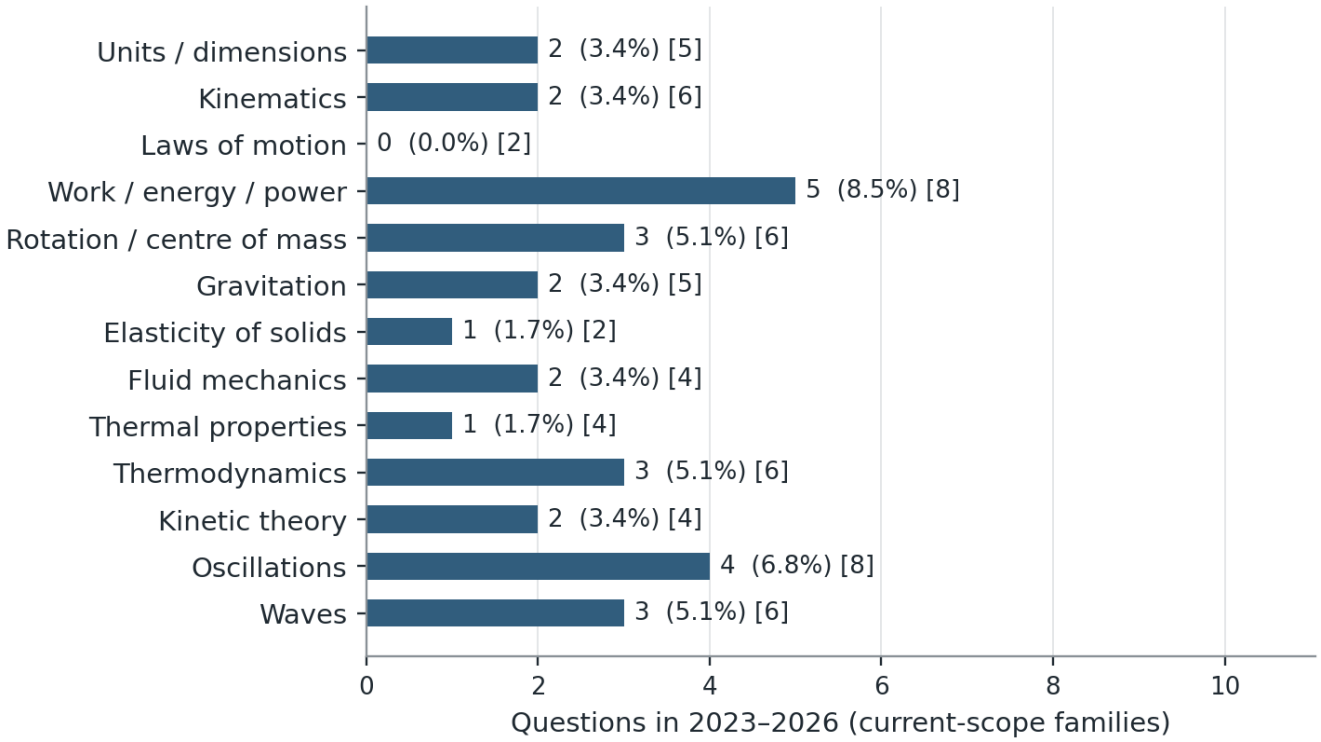
Three joint leaders: coordinate geometry, differentiation and integration have 6 questions each. Together they contribute 18 of 59 questions (30.5%); the rest of the syllabus contributes the other 69.5%.

Practise: domain restrictions, principal values, symmetry, root location and exact counting. In calculus, decide whether a transformation or a graph can simplify the problem before expanding algebra.

Limits, functions and probability span material taught across both school years. A chapter-family label is not a precise item-by-item Class 11/12 classification.

Physics: mechanics, heat and waves

Part 1 of 2 | Same denominator as the next page: 59 recent current-scope Physics questions.



Source: manual PYQ chapter ledger [S2]. Recent = 2023–2026. Shares use 59 current-scope questions in Physics; each question has one primary chapter. Bracketed counts [ ] show all acquired years for that family.

What repeats: work/energy has 5 questions, oscillations 4, and rotation, thermodynamics and waves 3 each. Energy conservation and force reasoning also appear as secondary tools in other chapters.

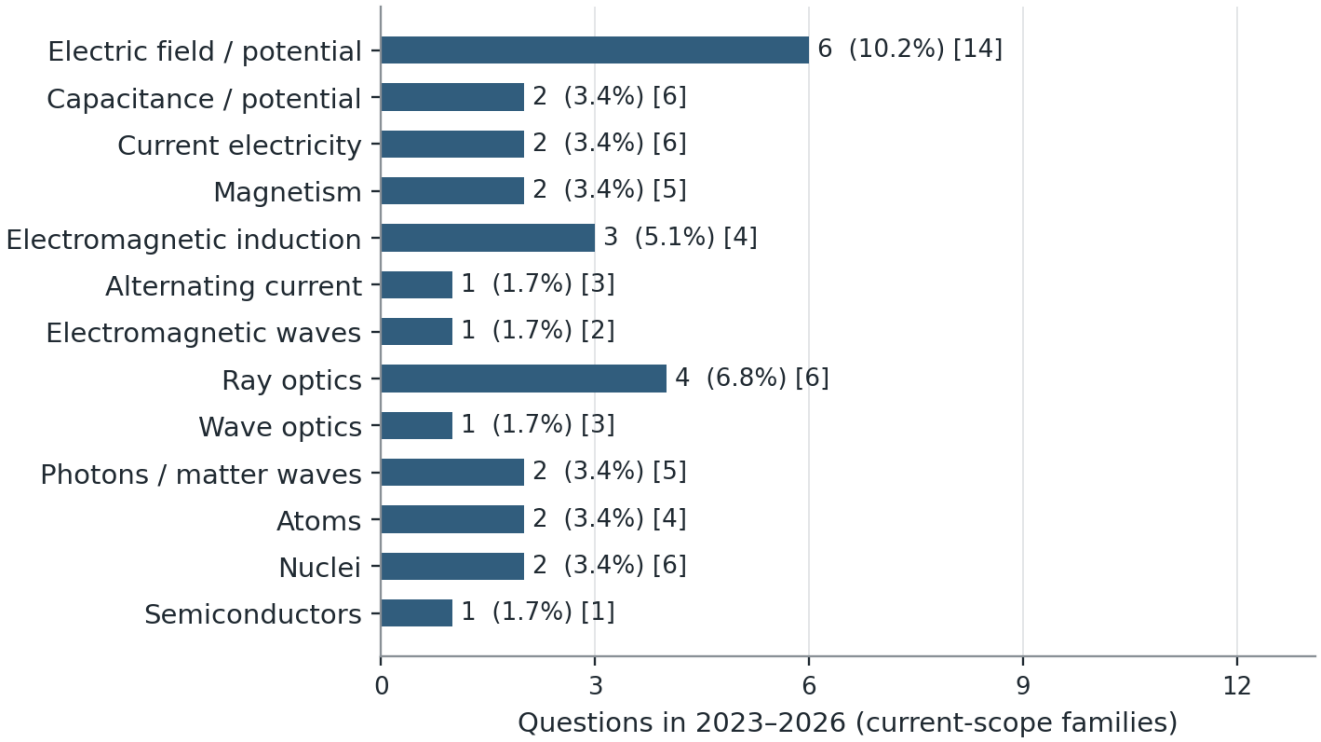
Do not confuse a zero with an excluded chapter

No recent question was assigned primarily to Laws of Motion in this four-paper window. That does not remove Newton’s laws from the syllabus or mean they were never used inside another problem.

Practise: drawing a force model, choosing a conservation law, distinguishing distance from displacement and linking an equation to a graph.

Physics: electricity, optics and modern topics

Part 2 of 2 | The shares on both Physics pages add to 100%, apart from rounding.



Source: manual PYQ chapter ledger [S2]. Recent = 2023–2026. Shares use 59 current-scope questions in Physics; each question has one primary chapter. Bracketed counts [ ] show all acquired years for that family.

What repeats: electric field/potential reasoning has 6 questions; ray optics 4; induction 3. Atoms, nuclei, matter waves and electronics appear across the window even when each has a smaller count.

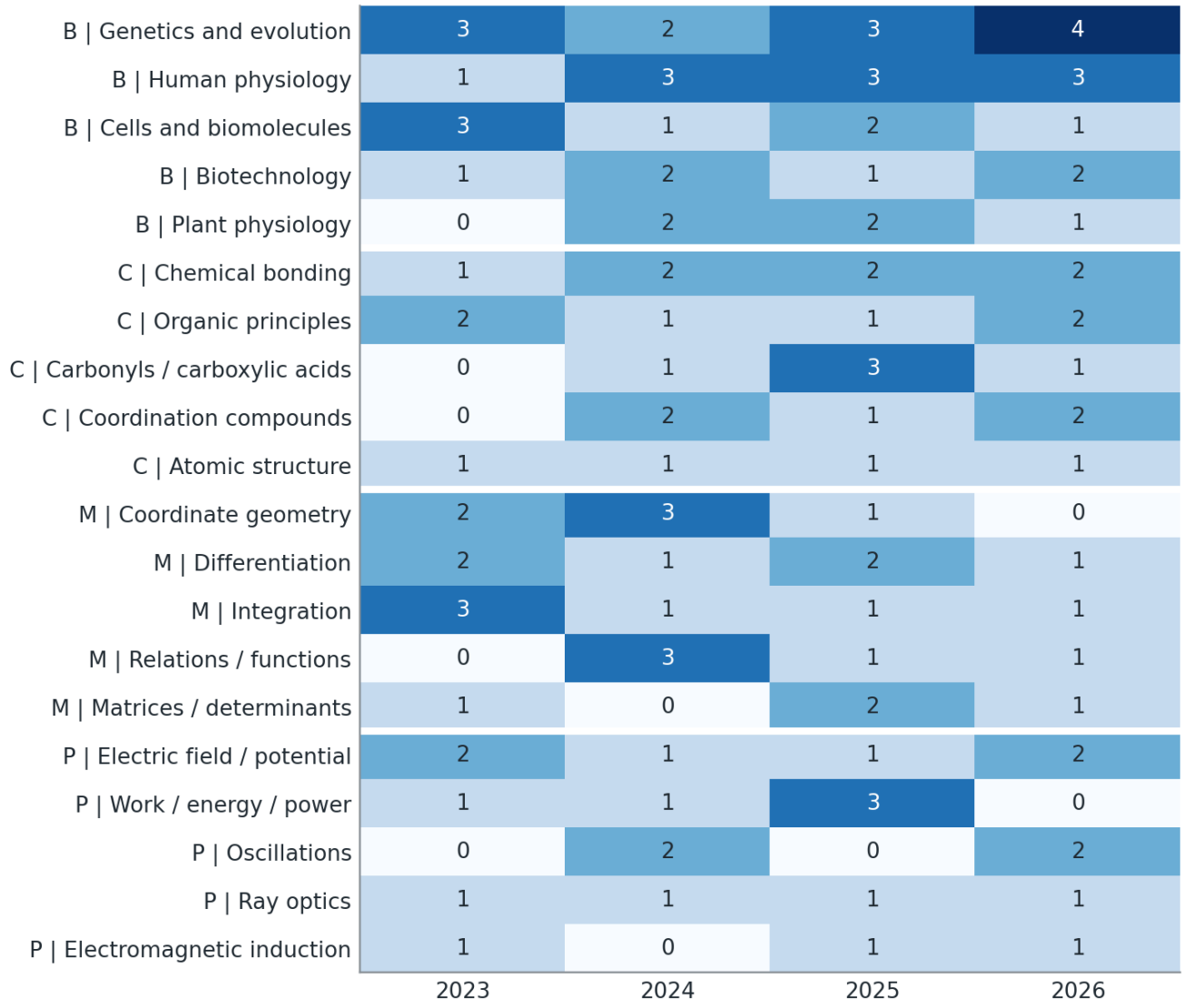
Treat figures as data

In a circuit, identify which nodes are connected before using a formula. In optics, check the medium, viewing direction and approximation. For a field graph, the sign of an area can decide whether kinetic energy rises or falls.

Electric field/potential and capacitance/potential are separate primary families in this taxonomy. Each question is placed in only one, so their displayed counts do not double-count items.

A frequent topic can still vary by year

Five leading families per subject, selected by recent count; ties use alphabetical chapter order.



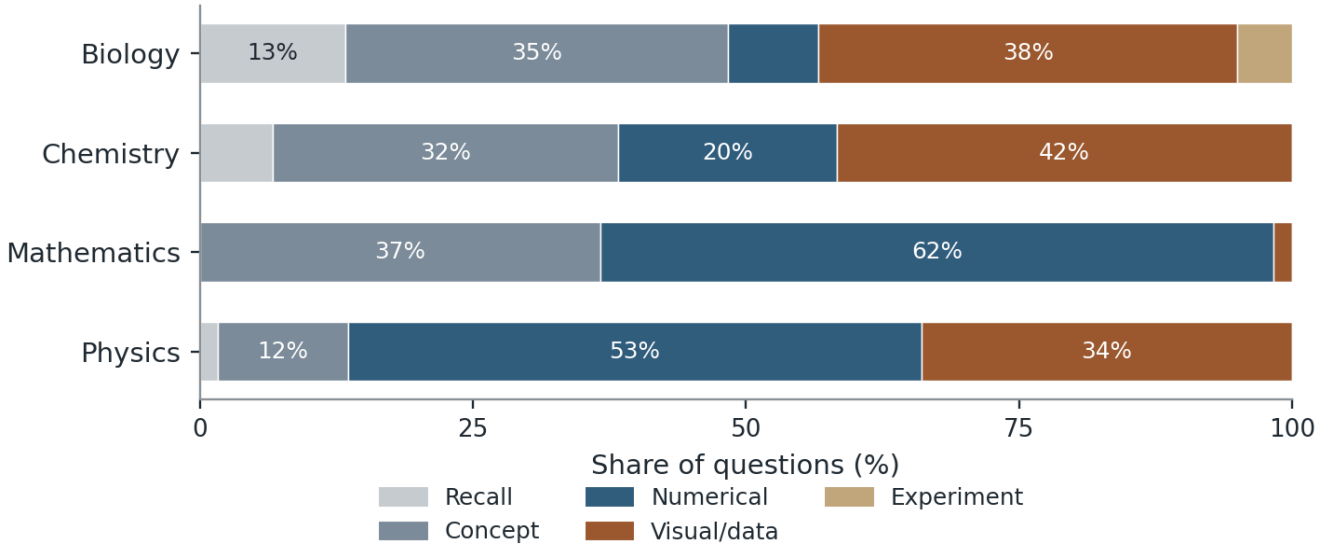
Each cell is an actual primary-topic count in that year [S2]. B/C/M/P mean Biology/Chemistry/Mathematics/Physics. The 2025 Physics source has 14 questions. Zero means no acquired item was tagged to that family that year.

Read across a row to see repetition and gaps. A chapter can lead the four-year total without being equally common every year. The dark cells show concentration in this sample, not a higher probability for next year.

Preparation rule: revisit recurring families, then rotate through the remaining syllabus. Do not try to turn a four-year heatmap into an exact next-paper forecast.

IAT asks for more than recall

Dominant task = the main kind of thinking needed. One question receives one task label.



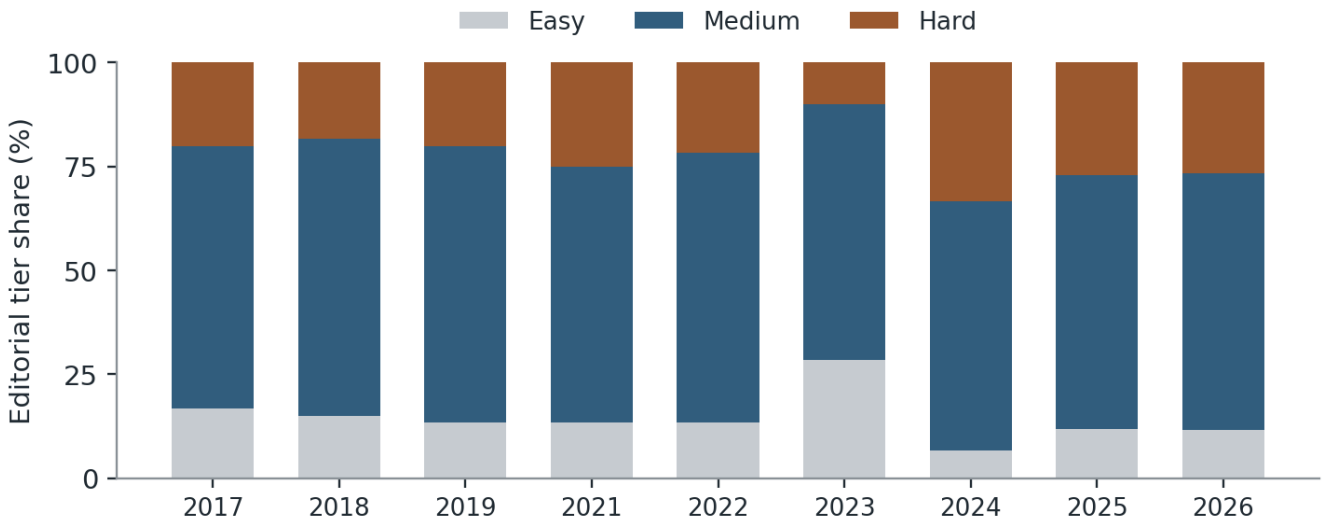
Recent unfiltered window: B/C/M n=60 each; P n=59. Visual/data combines diagrams/structures, graphs and tables. These task categories include scope-review items; chapter charts use filtered denominators.

Subject	What the pattern suggests
Biology	Compare explanations, interpret a cross or experiment, and use data to test a mechanism.
Chemistry	Read structures as well as words. A formula typed in a stem is not automatically a structure-reading task.
Mathematics	Work through cases and constraints, not only routine substitution. The numerical/concept boundary is editorial.
Physics	Model the situation and read figures, then calculate with signs and units.

Figure task totals: 23/60 Biology, 25/60 Chemistry, 1/60 Mathematics and 20/59 Physics. These include tables and molecular structures, not just separate image files.

Difficulty is an estimate, not a measured result

The labels below are editorial judgements about steps and reasoning, not student success rates.



All nine acquired papers; the 2025 denominator is 59. The years are separate categories: the missing 2020 paper is not interpolated. Source: manual difficulty tags [S2].

Recent window	Easy	Medium	Hard
Biology	12	39	9
Chemistry	9	38	13
Mathematics	9	38	13
Physics	5	31	23

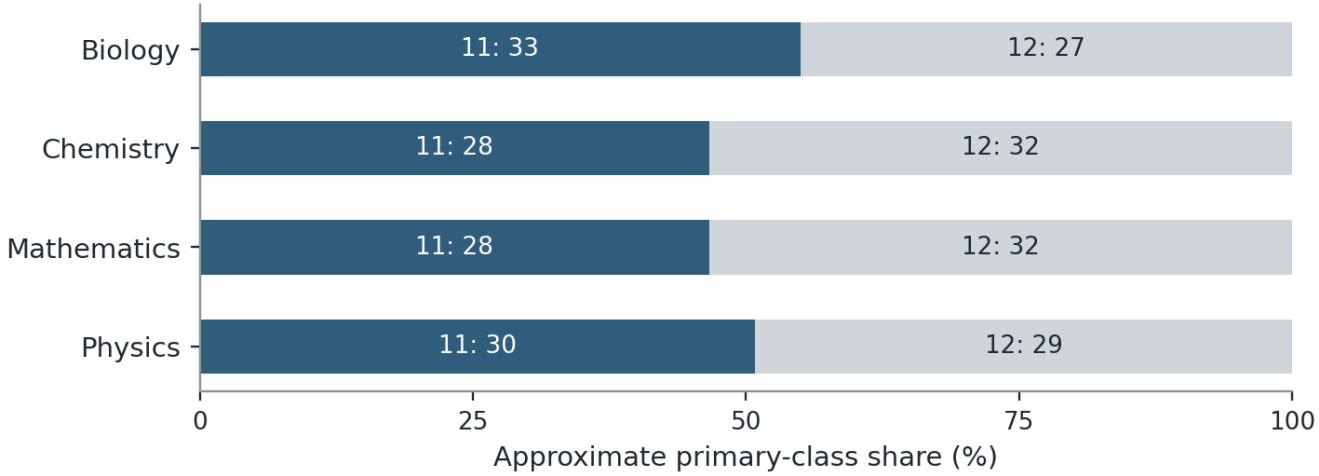
Physics has more hard tags under this rubric. That is a signal to practise modelling and longer reasoning, not evidence that a future paper must be harder or that a particular score will earn a rank.

Working label	A useful way to recognise it
Easy	One familiar idea and a short route
Medium	Linked ideas, cases or two to three substantive steps
Hard	Integration, a less obvious route or a longer derivation

The mock-series target of 9 easy / 39 medium / 12 hard is an OurMocks design rule. It is not an official IAT quota or a fit to every paper.

Prepare both school years; check the current syllabus

The Class 11/12 split below is approximate because labels are assigned at family level.



Recent, unfiltered questions. Numbers inside bars are counts. Some Maths families span both years, so do not use these bars as exact chapter quotas. Source: taxonomy-derived primary-class tags [S2].

Old papers are valuable, but old scope can differ

Across all years, **22 questions** are flagged as historical or needing scope review. Eight of these are in the recent window, all in 2023. The chapter charts exclude those recent review flags.

Subject	Examples that need scope checking
Biology	Transport/mineral nutrition, digestion and environmental issues; some nitrogen-fixation content can overlap retained microbes topics.
Chemistry	Detailed p-block, gases, solid state, polymers, metallurgy and surface chemistry.
Mathematics	Mathematical induction; linear programming is named only in an umbrella heading in the stored syllabus.
Physics	Logic gates and Zener-diode detail.

Use the latest official list: a question appearing in an old paper is not enough to prove it belongs in the target-year mock. Conversely, a low recent count does not exclude a listed topic.

Turn the patterns into a study plan

These are preparation priorities, not predicted questions or guaranteed chapter counts.

Subject	Revisit often	Keep in rotation
Biology	Genetics, cells, human physiology, biotechnology	Diversity, structural biology, plants, reproduction, ecology, welfare
Chemistry	Bonding, organic principles, coordination, carbonyl chemistry	All physical families plus retained inorganic and organic chapters
Mathematics	Calculus, functions, coordinate geometry, matrices, probability	Counting, binomial, sequences, complex numbers, vectors, 3D, statistics, trigonometry
Physics	Energy, electrostatics, oscillations, optics	Remaining mechanics, thermal, electricity, magnetism and modern topics

A repeatable practice cycle

• **Learn:** revise the NCERT idea and the conditions under which it applies.

• **Apply:** solve a mix of conceptual, numerical and visual questions without looking at the explanation first.

• **Review:** write down whether the error came from a concept, representation, calculation, assumption or time decision.

• **Return:** solve a different problem on the same idea after a gap; do not rely on remembering the option letter.

During mock review, explain why the other three options fail. If you guessed correctly, still record the missing reasoning. A correct mark can hide a weak method.

No reliable probability forecast, cut-off or rank model is built here. Such claims would need cleaner complete data, independent tagging and testing on unseen papers.

Keep an error log and know the sources

Use the evidence to guide practice, not to narrow your preparation to a few chapters.

Question / topic	What went wrong?	What I will change
Example: field graph	Used unsigned area	Write the field sign before integration
_____	_____	_____
_____	_____	_____

Small glossary

PYQ: previous-year question. **Weightage:** counted share of a defined set. **n:** number of questions. **Primary chapter:** the one main family assigned to an item. **Editorial difficulty:** a reviewer's estimate, not a measured pass rate.

Sources and reproducibility

[S1] Acquired reference collection: PYQ/IAT/manifest.json and SOURCES.md. Stored PDFs are labelled mirror copies; original pages remain authoritative when OCR is unclear.

[S2] Manual ledger, tags, taxonomy and metrics in docs/blueprints/iat_pyq_*.json / .txt; complete all-year and recent chapter counts in iat_pyq_chapter_counts.md. Counts are reproducible with tools/iat/analyse_pyq.py. Chapter labels, styles and tiers can be reviewed and changed transparently.

[S3] Official 2026 exam page: iiseradmission.in/pages/admission/iat.html. Official Information Brochure: app.iiseradmission.in/assets/pdfs/IB_IAT_2026_ENG.pdf, sections 19 and 23; checked 29 October 2026. Rules are paraphrased, not reproduced as an official document.

[S4] Stored official syllabus: IISER/IISER_Aptitude_Test_Syllabus_2026.md, extracted from the accompanying PDF. No 2027 brochure is verified in this report.

Prepared with AI assistance from the project's evidence. Human review is appropriate for item-level chapter and difficulty judgements. Figures show counted data, not invented trends or student-performance statistics.